

TOPICAL REVIEW

Content Advisors: Keith Muir, MD, and Gustavo Saposnik, MD, MPH, PhD

Midlife as the Critical Window for Women's Stroke and Dementia Prevention: Pivotal Advances and Implementation Priorities

Aleksandra Pikula¹, MD; Sharon Iziduh², DrPH, MBBS; Emine Kocabas, MPH; Evy Reinders³, MD; Ana Claudia de Souza⁴, MD, PhD; Sanjula D. Singh⁵, MD, PhD, MSc

ABSTRACT: Women face a higher lifetime risk of stroke and dementia than men, yet prevention strategies are not optimally implemented for women. Midlife (ages 40–65) represents a critical yet underused life stage window for prevention. This period encompasses the menopausal transition, when hormonal, metabolic, vascular, and neurological changes converge. Despite this, women remain underrepresented in stroke trials, and sex-specific risk factors are poorly integrated into routine prevention. This review synthesizes the transformative advances in midlife women's stroke and vascular dementia prevention from 2015 to 2025, drawing on evidence from clinical guidelines, cohort studies, neuroimaging studies, randomized trials, and implementation research. It highlights research priorities and an implementation framework for the future. Six major advances were identified: first, the 2024 American Heart Association/American Stroke Association guidelines introduced a class I recommendation to screen for adverse pregnancy outcomes. Second, neuroimaging now demonstrates that adverse pregnancy outcomes are linked to vascular brain injury decades after they occur. Third, the 2020 American Heart Association statement established menopausal transition as a distinct cardiometabolic and neurobiological transition affecting vascular brain health. Fourth, menopause hormone therapy controversies have resolved through evidence supporting personalized, timing-based approaches. Fifth, inflammatory and lipid biomarkers measured in midlife can predict stroke risk years out. Sixth, postmenopausal women show greater vascular benefits from physical activity. Despite these advances, women do not routinely receive appropriate screening or prevention recommendations. These findings support a comprehensive midlife assessment that incorporates reproductive history, menopausal transition status, and novel biomarkers alongside traditional risk factors, and should be offered streamlined recommendations for lifestyle strategies and menopause hormone therapy when indicated. Closing the implementation gap will require health system adaptation, provider education, and embedded pragmatic trials to incorporate these advances into routine care.

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Key Words: incidence ■ neuroimaging ■ postmenopause ■ quality of life ■ risk factors

Women have a higher absolute burden of stroke than men globally¹ as well as a higher lifetime risk of stroke or dementia.² In midlife, one in 2 women and one in 3 men will experience stroke or dementia.^{3,4} At age 65, a woman's remaining lifetime risk of developing either stroke or dementia is 50%, compared with 33% for men.⁴ Beyond incidence, women have higher 1-year mortality rates, greater disability, poorer functional recovery, and poorer quality of life after stroke.^{4–6}

Why Midlife Matters

Midlife (ages 40–65)⁷ encompasses significant sex-specific biological changes during the menopausal transition (MT), when hormonal, metabolic, vascular, and neurobiological systems undergo substantial shifts.⁸ As such, MT represents a critical window for implementing early intervention strategies to reduce stroke and vascular dementia risk.⁸ Women spend 30 to 35 years, ~40%

Correspondence to: Aleksandra Pikula, MD, Department of Medicine, Division of Neurology, University of Toronto, JS Sonshine Centre, Krembil Brain Institute, University Health Network, 399 Bathurst St, 5W-432, Toronto, ON, M5T 2S8, Canada. Email aleksandra.pikula@uhn.ca
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of their lives, in the postmenopausal stage,⁸ indicating that interventions during MT can influence decades of subsequent health outcomes. Yet women in midlife are underrepresented in research and clinical care.^{9–11} Though women have over 50% disease burden compared with men, they account for only 37% of stroke trial participants, with no substantial improvement over 3 decades.⁹ Furthermore, only about one-third of studies report sex-stratified outcomes.^{9,10}

Female-Specific Factors Influencing Midlife Stroke and Dementia Risk

Adverse pregnancy outcomes (APOs), including hypertensive disorders of pregnancy (5%–10% of all pregnancies), gestational diabetes (6%–9%), and preterm delivery (10%–12%), are residual vascular risks carried over to midlife and have been consistently linked to elevated long-term risks of cerebrovascular and cardiovascular disease.^{11–14} Notably, these complications disproportionately affect Black, Indigenous, and socioeconomically disadvantaged women, who experience both higher APOs rates and greater barriers to postpartum and midlife preventive care, compounding lifetime cerebrovascular risk.^{8,14} These associations persist for decades and likely reflect enduring endothelial dysfunction, inflammation, and other biological pathways.¹⁵ In a 2025 Swedish nationwide study of over 2.2 million women followed for up to 46 years, all major APOs were independently associated with increased stroke risk.^{12–14}

During midlife, MT triggers significant cardiometabolic and neurobiological changes that are partly independent of chronological aging.^{16–18} Clinically, vasomotor symptoms (VMS) are not only highly prevalent, but are also associated with subclinical cardiovascular dysfunction,¹⁹ and a transient decline in verbal memory that tends to improve as women transition through menopause.²⁰ Sleep disruption affects 40% to 60% of midlife women and has been linked to increased dementia risk.²¹ Interestingly, the odds of experiencing depressive symptoms are significantly higher during perimenopause and postmenopause compared with premenopause,^{22,23} and evidence confirms depression as a significant risk factor for dementia.²⁴ Figure [A] illustrates the life course of women, exposures at each stage, and neurovascular outcomes.

Importantly, converging neuroimaging and mechanistic evidence shows that women with prior hypertensive disorders of pregnancy exhibit greater cerebral small vessel disease even 20 to 30 years later,²⁵ and that midlife hormonal and cardiometabolic transitions are accompanied by measurable alterations in cerebrovascular structure, perfusion, and neural network integrity,²⁶ highlighting midlife as a biologically active window for targeted prevention.

Here, we synthesize key advances (2015–2025) in midlife women's prevention of stroke and vascular

dementia, spanning guidelines, cohort studies and neuroimaging evidence, randomized trials, and implementation research, and we propose priorities and an implementation framework. This topic review offers a distinct contribution by integrating evidence streams that are rarely synthesized together, including epidemiology, neuroimaging, sex-specific physiology, biomarkers, and implementation science, to create a unified, midlife-focused prevention model for women. Unlike prior reviews that address pregnancy complications, menopause, or cardiovascular risk in isolation, this article connects these domains across the life course and identifies midlife as the biological and clinical inflection point where prevention can have the greatest long-term impact. By linking the past decade's transformative advances with recommendations and an implementation framework, this synthesis moves the field beyond describing risk toward actionable strategies that clinicians, health systems, and policymakers can adopt now to reduce women's lifetime stroke and dementia burden.

The Past Decade's Transformation

Over the past decade, women's midlife brain health has shifted from neglected to a recognized priority. Six major advances made in the past decade are showcased here. These advances collectively mark a shift in how midlife women's brain and vascular health should be conceptualized and managed. Table 1 summarizes the advances and challenges with actions to address the challenges.

Advance 1: Guideline Evolution—From Recognition to Clinical Mandate

Recent guidelines now have formal clinical mandates regarding screening for APOs. The 2014 AHA/ASA guidelines for primary stroke prevention were the first to comprehensively highlight female-specific factors such as APOs as modifiers of cerebrovascular risk, but their recommendations were conditional.²⁷ A decade later, the 2024 AHA/ASA guideline for primary stroke prevention introduced a transformative class I recommendation: In adults, screening for a history of certain APOs, including hypertensive disorders of pregnancy, preterm birth, gestational diabetes, and placental disorders, is recommended as part of stroke prevention strategies in females.¹⁴ Evidence accumulated over 2 decades demonstrates that hypertensive disorders of pregnancy confer ≈ 1.8 -fold higher stroke risk,¹³ gestational diabetes increases type 2 diabetes risk 7-fold,²⁸ and preterm delivery is associated with a 1.5 to 2-fold increase in cardiovascular disease.²⁹ Mechanistic studies further reveal persistent endothelial dysfunction and microvascular injury.^{30–32} International guidelines are

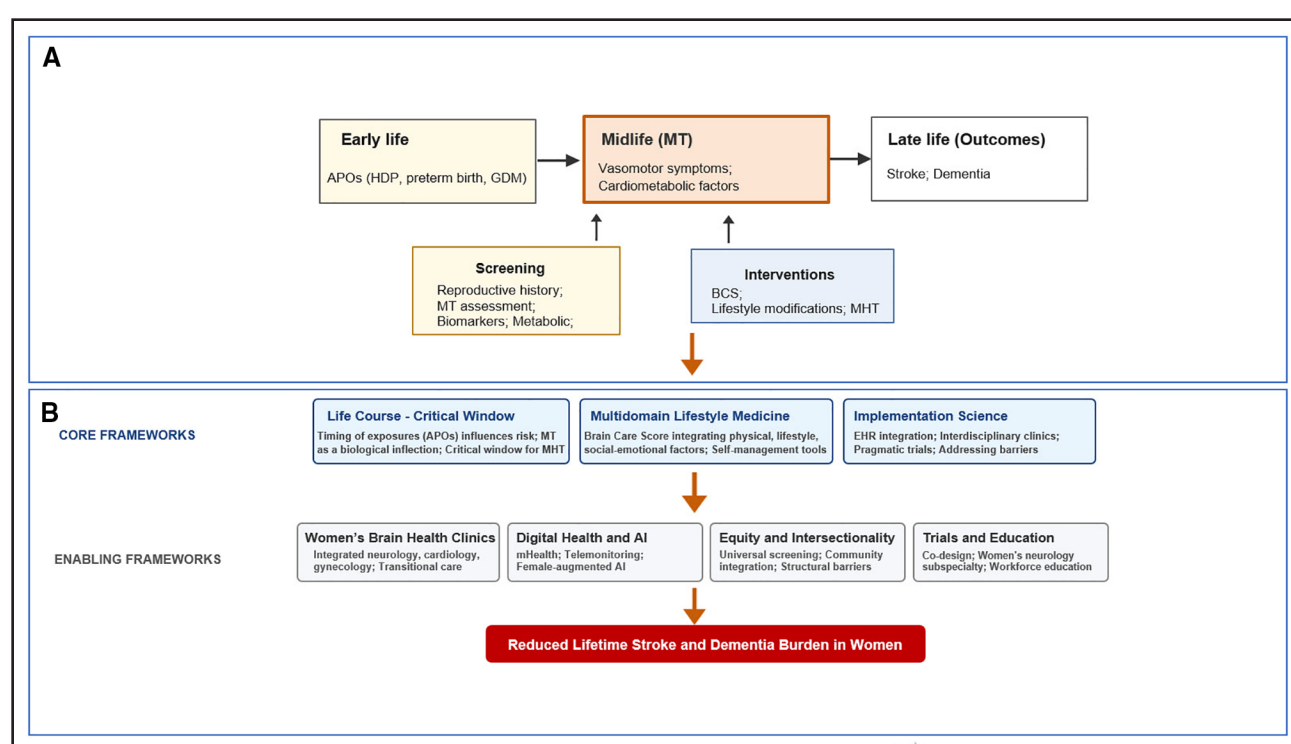


Figure. Integrated life course model and evidence frameworks for midlife women's stroke and dementia prevention.

A, Illustrates the life course trajectory from early life exposures through the menopause transition to late-life outcomes, highlighting critical windows for screening and intervention during midlife. **B**, Presents the core and enabling frameworks necessary to translate evidence into clinical practice, ultimately aiming to reduce lifetime stroke and dementia burden in women. AI indicates artificial intelligence; APOs, adverse pregnancy outcomes; BCS, Brain Care Score; EHR, electronic health record; GDM, gestational diabetes; HDP, hypertensive disorders of pregnancy; MHT, menopause hormone therapy; and MT, menopause transition.

now highlighting the role of APOs in stroke and dementia. Canadian Stroke Best Practice Recommendations emphasized APOs as critical determinants of long-term stroke risk.³³ The European Stroke Organization guidelines highlighted pregnancy, postpartum, and menopause as vulnerable periods, and the 2025 European Society of Cardiology guidelines included a dedicated chapter on long-term cardiovascular and stroke consequences of APOs.^{34,35}

Advance 2: Mechanistic Breakthroughs—From Epidemiology to Biology

Although guidelines now highlight the importance of reproductive history, mechanistic studies have provided evidence that APOs can precede cerebrovascular injury. Large registry analyses, including Crump's Swedish sibling study, confirmed that APOs predict stroke decades later,¹² and neuroimaging has been pivotal in reinforcing these epidemiological findings. The CARDIA prospective cohort followed women for 30 years from young adulthood and found that those with prior preeclampsia or gestational HTN had significantly greater white matter hyperintensities burden at midlife, along with accelerated cognitive decline.³⁶ At the population level, white matter hyperintensities confer a 2- to 3-fold increased

risk of dementia and stroke,³⁷ while silent brain infarcts more than double dementia risk.³⁸ Fluorodeoxyglucose-positron emission tomography studies demonstrated reduced glucose metabolism in Alzheimer disease-vulnerable regions (posterior cingulate, precuneus, parietotemporal cortex) during menopause.³⁹ Longitudinal imaging further reveals an Alzheimer disease-like bioenergetic phenotype characterized by mitochondrial dysfunction and white matter catabolism during peri- and postmenopause.⁴⁰ White matter hyperintensity volume increases across early postmenopause, consistent with microvascular injury.⁴¹ These findings together reflect underlying biological mechanisms, including lasting endothelial dysfunction,^{30–32} chronic inflammation,³¹ metabolic dysregulation,^{11,42} and adverse cardiometabolic profiles that persist into midlife.^{11,42,43}

Advance 3: Recognition of Menopause as Cardiometabolic and Neurobiological Transition

Although APOs predispose women to vascular vulnerabilities, the MT marks another critical point when traditional risks are emerging. The 2020 AHA Scientific Statement on Menopause Transition and Cardiovascular Disease Risk signified institutional recognition that menopause is a natural biological transition that warrants

Table 1. Summary of Advances, Challenges, and Actions

	Advances	Details	Challenges	Actions
1	Guidelines evolution	2024 AHA/ASA class I recommendation to screen for APOs	Uptake of reproductive history screening is low: <30% of eligible women are assessed.	Include screening for APOs in routine care Have APO screening embedded in EHR Train providers on screening for APOs
2	Mechanistic breakthroughs	Neuroimaging linked APOs to brain injury WMH and silent infarcts increase stroke and dementia risk	Lack of awareness of the long-term effects of APOs on the brain Suboptimal integration into risk models	Educate providers and patients Include APOs in cerebrovascular risk screening tools
3	Recognition of menopause as a cardiometabolic and neurobiological transition	MT is associated with cardiometabolic and neurobiological shifts and cognitive impairment	MT is not routinely assessed Cognitive symptoms are usually dismissed Poor timing of the intervention	Screen for MT status Interventions should be personalized Monitor cognitive and vascular markers
4	Menopause hormone therapy: resolving 2 decades of controversy	WHI reinterpretation and new trials Timing of administration validated Transdermal estradiol is safer Support the use and benefits of MHT	Fear that resulted from the initial WHI interpretation still persists Low MHT prescription rates Route and timing are not individualized	Educate providers and patients on timing and route Update protocols for MHT use
5	Integrated lifetime risk assessment in midlife women	hsCRP, LDL, and Lp(a) predict 30-year stroke risk Lp(a) remains stable across the lifespan and can be targeted for risk reduction	Biomarkers are not included in standard cardiovascular risk calculators Lp(a) testing is rarely used	Biomarkers should be screened for in midlife Cardiovascular risk calculators should be updated to include biomarkers
6	Sex-specific exercise physiology and lifestyle precision	Physical activity reduces stroke risk Effects of MT are modified by resistance training	Lifestyle intervention for stroke prevention is underused	Promote lifestyle medicine pillars

AHA/ASA indicates American Heart Association/American Stroke Association; APO, adverse pregnancy outcome; EHR, electronic health record; hsCRP, high-sensitivity C-reactive protein; LDL, low-density lipoprotein; Lp(a), lipoprotein (a); MHT, menopause hormone therapy; MT, menopausal transition; and WHI, Women's Health Initiative.

clinical attention.⁸ During MT, visceral adiposity, insulin resistance, low-density lipoprotein (LDL) cholesterol, and systolic blood pressure increase while endothelial function deteriorates. All these are individual risk factors for stroke or dementia.^{8,16–18,44–50}

Neurobiological data provide further rationale for midlife action. Recent multimodal imaging studies report reductions in brain glucose metabolism and increased β -amyloid deposition during perimenopause.^{39,51} These findings suggest that the preclinical Alzheimer's disease phase coincides with the endocrine transition of perimenopause, with metabolic decline correlating with estradiol reductions.^{19,52,53} Structural magnetic resonance imaging studies reveal corresponding hippocampal volume loss and accumulating white matter hyperintensities as risks of dementia and stroke, respectively.^{26,41} More importantly, HTN further compounds this vulnerability, accelerating regional brain volume decline over time.⁵⁴ The ARIC study demonstrated that midlife vascular risk factors (including HTN), particularly in earlier midlife (ages 44–54), were strongly associated with 25-year dementia incidence, with weaker effects when risk factors developed later in life.⁵⁵

However, cognitive symptoms during MT require careful framing. The SWAN study found that subjective

forgetfulness was common.⁵⁶ However, when put into a more objective context, the SWAN cognitive substudy demonstrates that late perimenopausal women did not show the expected improvement in processing speed that was evident in premenopausal, early perimenopausal, and early postmenopausal women. This pattern suggests a transient window of cognitive vulnerability specific to late perimenopause that, in most women, resolves with the transition to postmenopause.⁵⁷ These findings collectively identify late perimenopause as a critical therapeutic window when interventions targeting hormonal, vascular, or metabolic pathways may be most effective before cognitive changes become persistent. The critical window hypothesis suggests that the vascular and metabolic effects of menopause hormone therapy (MHT) may depend on the timing of initiation; however, evidence to date does not demonstrate sustained cognitive benefit. Importantly, the Canadian Longitudinal Study on Aging showed that earlier menopause was associated with worse cognitive outcomes, indicating that endocrine and vascular processes act synergistically, with the MT amplifying preexisting vulnerabilities.⁵⁸ Optimal strategies for this critical window, including hormone therapy, vascular risk modification, and lifestyle-based interventions, remain an active area of investigation.

Advance 4: Menopause Hormone Therapy Resolving 2 Decades of Controversy

MHT remains one of the most contested topics in women's preventive health. Over the past decade, evidence has clarified core sources of controversy, particularly timing and route of administration, and baseline vascular risks, shifting practice toward individualized, risk-stratified decision-making. The primary clinical indication for MHT is the treatment of VMS, which affects up to 80% of women and impacts their quality of life.⁵⁹ Therefore, the severity of VMS should be considered when evaluating the risk-benefit ratio of MHT, as women with moderate to severe symptoms may experience greater benefits.⁵⁹

The WHI Reinterpretation

The 2002 Women's Health Initiative (WHI) trial reported increased risk of cardiovascular events, stroke, venous thromboembolism, and breast cancer with oral conjugated equine estrogens (CEE) plus medroxyprogesterone acetate (MPA).⁶⁰ In response, MHT prescriptions fell by ≈80%.⁶¹ Critical nuances were lost for a few decades: the trial enrolled older women (mean age 63), averaging a decade past menopause, used a specific oral formulation, and excluded women with bothersome VMS, subsequently leaving many symptomatic women untreated.^{60,61}

The Timing Hypothesis and ELITE Trial

In 2013, age-stratified WHI reanalysis revealed that cardiovascular outcomes depend critically on when MHT was started relative to menopause onset.⁶² Among women who began MHT within 10 years of their final menstrual period, coronary artery disease was reduced by 24% (hazard ratio [HR], 0.76). Women who started therapy 20+ years after menopause experienced a 28% increase in coronary artery disease events (HR, 1.28; *P* for trend=0.02).⁶² The ELITE trial provided direct experimental confirmation showing that estradiol reduced carotid intima-media thickness progression when initiated within 6 years of menopause but had no effect when started 10+ years later; however, coronary artery calcium progression did not differ between the early and late initiation groups.⁶³ In parallel, a meta-analysis confirmed that early menopause (<45 years) increased coronary artery disease risk by 50%.⁶⁴ Yet, unlike the coronary findings, WHI consistently demonstrated that increased ischemic stroke risk with MHT was not modified by initiation timing.⁶²

Now, how does the timing of MHT initiation relative to menopause influence cardiovascular or cerebrovascular benefit versus risk? Mechanistic rationale behind the timing hypothesis emerged, where the so-called healthy cell premise centered on the vascular substrate.⁶³ In early

postmenopausal years, healthy endothelium responds to estrogen with vasodilation and anti-inflammatory signaling, but once atherosclerosis is established, estrogen may destabilize vulnerable plaques.⁶³ This timing dependence reflects shifts in ER α (estrogen receptor expression), which predominates in healthy endothelium and drives nitric oxide production, and becomes dysregulated in atherosclerotic vessels.⁶⁵ Concurrently, estrogen withdrawal during MT disinhibits microglial activation and increases proinflammatory cytokine production, contributing to cognitive vulnerability during late perimenopause.²⁶ Route also matters, as oral estrogens undergo hepatic first-pass metabolism, increasing clotting factors and predominantly converting to estrone, whereas transdermal estradiol bypasses the liver, maintains stable estradiol levels, and carries a substantially lower thromboembolism risk.^{63,66,67}

Regarding cerebrovascular risk, the WHI randomized controlled trial demonstrated that both standard-dose oral CEE+MPA and CEE alone increased ischemic stroke risk by 30% to 40% regardless of age at initiation.^{60,62} More broadly, cardiovascular risk was also elevated with oral CEE+MPA and tibolone in the 2024 Swedish nationwide emulated target trial of 919614 women, whereas transdermal estradiol regimens (≤ 50 $\mu\text{g}/\text{d}$) demonstrated a favorable safety profile.⁶⁸ The absence of elevated stroke risk with transdermal estradiol ≤ 50 $\mu\text{g}/\text{d}$ is further supported by observational evidence from a population-based matched case-control study, which provides reassurance but does not carry the inferential strength of an RCT.^{67,68}

With respect to cognitive outcomes, the WHI Memory Study (women aged 65 to 79 years) showed that CEE+MPA doubled the risk of probable dementia (HR, 2.05 [95% CI, 1.21–3.48]).⁶⁹ In contrast, trials enrolling younger women within a few years of menopause onset (KEEPS Cog, ELITE cognitive ancillary) found no cognitive benefit or harm with 4 to 5 years of treatment using transdermal estradiol or lower-dose oral estrogen paired with micronized progesterone (oral or vaginal, respectively).^{70,71} Overall, current evidence indicates that the cognitive and cerebrovascular effects of MHT vary by factors, such as time since menopause and route of administration. Because MHT is used primarily for VMS management, its severity remains central to treatment decisions. Baseline cardiovascular risk also influences safety. Oral CEE, with or without MPA, has been consistently associated with increased ischemic stroke risk across both randomized trials (WHI)^{60,62} and observational studies (eg, Nurses' Health Study).^{8,14} In contrast, observational data suggest a more favorable cerebrovascular profile with transdermal estradiol.^{67,68} Taken together, these considerations support individualized MHT decisions based on VMS burden, time since menopause, route of administration, and cardiovascular risk at baseline.

Advance 5: Integrated Lifetime Risk Assessment in Midlife Women

Most cardiovascular risk prediction tools were designed for 5- to 10-year horizons and may be poorly aligned with midlife women's biology and life-course trajectories.⁷² Recent long-term cohort evidence reframes midlife as the optimal time to measure biomarkers that forecast risk decades later.^{73–75}

The Women's Health Study (27 939 women; median 30-year follow-up) demonstrated that biomarkers measured in midlife women predict ischemic stroke 3 decades later.⁷³ Three biomarkers, such as hsCRP (high-sensitivity C-reactive protein), LDL cholesterol, and Lp (a) (lipoprotein [a]), individually predicted 15% to 30% higher 30-year stroke risk (HRs, 1.15–1.33 per SD increase). Women with multiple elevated biomarkers had 2.4-fold increased stroke risk (HR, 2.41 [95% CI, 1.93–3.01]).⁷³ These findings were also confirmed by the Independent EPIC-Norfolk validation.⁷⁴

The novelty lies not in the biomarkers themselves, but in recognizing that midlife offers an optimal window for comprehensive cardiovascular risk stratification in women.^{73,74,76} The 3 biomarkers capture distinct pathophysiologic mechanisms: hsCRP reflects chronic systemic inflammation, LDL remains the primary atherogenic lipoprotein, and Lp(a) is a genetically determined LDL variant with prothrombotic properties.^{16,73–77} Lp(a) is especially relevant for lifetime stratification because ≈20% of women have Lp(a) > 50 mg/dL.⁷⁵ Unlike LDL, Lp(a) remains stable across the lifespan, making midlife measurement definitive for lifetime risk stratification.^{75,78}

Clinical Integration Priority

Comprehensive reproductive history documentation is recommended per 2024 AHA/ASA guidelines.¹⁴ Women in their 40s with a combined APOs history and multiple elevated biomarkers face substantially elevated cumulative lifetime stroke risk, well above population averages, yet the risk is undetected by traditional 10-year calculators.^{13,72,73,77} Therefore, standard midlife assessment should include 1 × hsCRP, Lp(a) measurement (ages 40–55) and hemoglobin A1c for metabolic screening. Hemoglobin A1c screening is particularly important for women with a history of gestational diabetes, who face a 7-fold increased risk of developing type 2 diabetes.^{28,29} Importantly, the 2025 ACC Scientific Statement endorses hsCRP and Lp(a) as integral to contemporary risk assessment.⁷⁶ Lp(a) measurement rates, however, remain below 20% even in specialty practices,^{75,78,79} and most cardiovascular risk calculators still exclude pregnancy complications.^{11,14}

Advance 6: Sex-Specific Exercise Physiology and Lifestyle Precision

Lastly, lifestyle medicine has emerged as having a key role in modifying stroke risk and sex-specific exercise physiology, providing benefits for women in midlife.

Physical Activity

Large prospective cohorts, including the Nurses' Health Study, demonstrate that physical activity reduces stroke risk in women, with benefits evident across menopausal stages.⁸⁰ Exercise enhances endothelial shear stress and nitric oxide bioavailability, mechanisms that parallel estrogen's vascular effects, explaining why postmenopausal women respond robustly to regular activity.⁸¹ Guideline-consistent thresholds (150–300 minutes/wk moderate intensity or 75–150 minutes/wk vigorous intensity) are associated with cerebrovascular and cardiometabolic benefits.¹⁴

Importantly, combined aerobic and resistance training provides optimal benefits for menopausal health.⁸² Resistance training 2 to 3× weekly preserves lean mass, improves insulin sensitivity, and counters visceral adiposity.⁸ Longitudinal midlife women's cohorts confirm accelerated muscle loss during the transition,^{8,82} highlighting resistance training as essential for maintaining metabolic resilience.

Multidomain Lifestyle Approaches

In addition to physical activity, a Mediterranean-style diet was also associated with significantly lower coronary artery disease risk in women.⁸³ Smoking cessation reduces cardiovascular disease mortality by ≈35% to 40% within 5 years.⁸⁴ Sleep disturbance across MT predicts higher cardiometabolic risk in observational studies, though whether sleep-targeted interventions during MT reduce cerebrovascular events remains an important research gap.^{8,85} Social isolation and loneliness predict incident stroke, with stronger associations in women, but specific interventions for midlife are not yet offered.^{86,87}

FUTURE DIRECTIONS

The evidence—prospective cohorts, mechanistic validation, and RCTs for MHT—justifies acting now despite some clinical trial gaps. Optimizing brain health across all women's life stages is paramount, and the tools to do so exist today. Table 2 highlights the recommended core components for midlife women's stroke and dementia risk assessment.

Implementation Readiness

Building on the 6 advances identified in this article, Table 3 highlights which interventions are ready for immediate implementation in clinical practice and those still in development.

Evidence Frameworks for Future Practice and Embedded Research

In the next section, we briefly outline 3 core frameworks to guide future research efforts, along with 4 enabling frameworks as illustrated in Figure [B].

Table 2. Recommended Core Components to Consider for Midlife Stroke and Dementia Risk Assessment in Women

Component	Elements
1. Reproductive history	Hypertensive disorders of pregnancy
	Gestational diabetes
	Preterm birth
	Parity
	Age at menopause
2. Menopausal transition status	Stage (early peri, late peri, post)
	Vasomotor symptoms
	Sleep disturbance
	Mood symptoms
3. Cardiometabolic and vascular risk factors	Blood pressure
	Lipid panel
	Hemoglobin A1c
	BMI
4. Biomarkers	hsCRP
	Lp(a)
	LDL cholesterol
5. Lifestyle and behavioral factors	Physical activity
	Nutrition
	Restorative sleep
	Positive social connections
	Avoidance of risky substances
	Stress management
6. MHT eligibility and risk assessment	Symptom profile
	Time since menopause
	Baseline cardiovascular risk
	Thromboembolic risk
	Stroke risk considerations
	Route of administration

BMI indicates body mass index; hsCRP, high-sensitivity C-reactive protein; LDL, low-density lipoprotein; Lp(a), lipoprotein (a); and MHT, menopause hormone therapy.

Life Course Health Development Framework and the Concept of Critical Window

The life course health development framework explains how different environmental, physiological, behavioral, and psychological contexts influence risk profiles and long-term health development trajectories through life stages,^{88,89} and as such outlines a critical window for targeted preventive interventions.⁹⁰

Multidomain Lifestyle Medicine Framework

Midlife cardiovascular risk factors and lifestyle behaviors (exercise, diet, smoking) are associated with stroke and cognitive decline and dementia risk in observational studies of women.^{8,91} One promising tool within the multidomain lifestyle medicine framework is the Brain Care Score (BCS). The BCS, validated in 398990 UK Biobank participants, integrates 12 modifiable factors across

Table 3. Implementation Readiness of Prevention Components

Ready for implementation	Emerging
APO screening (2024 AHA/ASA class I)	Integration of reproductive history into EHR
MT status assessment	Digital tools for MT symptom tracking
Cardiometabolic status	RNA-based Lp(a)-lowering therapies
hsCRP and Lp(a) measurement in midlife	Neuroimaging markers (WMH trajectories, PET phenotypes)
Early, personalized MHT for women without contraindications	
Lifestyle interventions	

AHA/ASA indicates American Heart Association/American Stroke Association; APO, adverse pregnancy outcome; EHR, electronic health record; hsCRP, high-sensitivity C-reactive protein; Lp(a), lipoprotein (a); MHT, menopause hormone therapy; MT, menopausal transition; and WMH, white matter hyperintensities.

physical, lifestyle, and social-emotional domains, offering a patient-facing self-management tool that also promotes behavioral change.⁹² Among adults under 50, each 5-point increase was associated with 59% lower dementia risk and 48% lower stroke risk. Validation among women also confirmed that a 5-point increase in BCS was associated with a 37% lower risk of cerebrovascular events over 22 years of follow-up.⁹³ This could further enhance precision for women and offer self-management tools to support ongoing primary prevention.

Implementation Science Framework

Knowledge translation requires integrating APOs screening, reproductive history documentation, and biomarker data into electronic health records; establishing interdisciplinary prevention clinics; and conducting pragmatic trials. Canadian Maternal Health Clinic models demonstrated the feasibility of postpartum screening, but also highlight substantial attrition driven by childcare, work, and risk-perception barriers.^{94,95} Pairing maternal follow-up with infant well-child visits may improve engagement but requires additional resources.⁹⁶

Midlife Women's Brain Health Clinics Framework

Midlife-focused prevention clinics integrating neurology, cardiology, gynecology, and primary care offer systematic reproductive history capture, enhanced biomarker assessment, and coordinated decision-making around MHT and other brain health preventive therapies. Such clinics, when additionally integrated with existing cardio-obstetrics programs, could support lifespan brain-health surveillance.

Digital Health and AI Framework

Digital technologies offer scalable solutions for stroke prevention among midlife women. Randomized mHealth interventions have demonstrated improvements in lifestyle behaviors and intermediate risk factors in high-risk populations.⁹⁷

Although mHealth platforms offering sex-specific content and digital coaching represent scalable tools for cardiometabolic and neurovascular risk management during high-risk postpartum and MT periods, their promise depends critically on female-augmented AI development.⁹⁸ Machine learning shows potential for delivering personalized women's health care, but can perpetuate historical sex and gender biases when trained on male-dominated data sets.⁹⁹ Training AI data sets must include reproductive history, hormonal cycles, and MT factors to ensure algorithms accurately predict risk and personalize interventions for women across their lifespan.

Equity and Intersectionality Framework

Black, Indigenous, and socioeconomically disadvantaged women experience a higher APO burden, delayed diagnosis, and reduced access to preventive interventions.^{8,14} Addressing these disparities requires universal screening protocols, language access, and community health worker programs.^{8,14,100} Simulation studies modeling hypothetical lifestyle strategies in midlife women further demonstrate the substantial stroke risk reductions achievable through scalable behavioral interventions.¹⁰¹ A 5-point higher BCS was linked to significantly lower risk of stroke, with a stronger association among Black individuals (HR, 0.47) compared with White individuals (HR, 0.75), suggesting greater potential benefit for Black populations.¹⁰² To ensure that midlife-focused prevention does not widen already existing disparities, accessibility must be prioritized in implementation.¹⁰⁰ Scalable models of care and composite measures like the BCS should be introduced into the community to increase access and promote equity.^{100,102}

Clinical Trials, Education, and Workforce Development Framework

Optimizing women's representation in prevention trials requires integration with obstetrics-gynecology practices and midlife clinics.⁹ Critically, codesign methodologies engaging midlife women as active partners in shaping research questions, trial protocols, and outcome measures can enhance both recruitment and the relevance of findings. Community partnerships, flexible scheduling, and pragmatic electronic health record-embedded designs further ensure women are central to evidence generation.^{8,14} The emerging subspecialty of women's neurology provides a core curriculum and infrastructure to integrate this evidence into clinical practice.^{8,9,103} Equally important, it is training a new generation of faculty equipped to support novel care models, foster interdisciplinary collaboration, and drive a paradigm shift in women's brain health research.

CONCLUSIONS

Midlife is the convergence point for women's vascular brain health. The past decade has delivered actionable

advances, yet translation into routine care remains uneven.

A practical path forward is to operationalize midlife women's prevention through (1) integration of education for patients and providers about midlife risks and potential interventions; (2) standardized reproductive history capture and APO screening; (3) routine MT staging and symptom-informed risk management; (4) selective biomarker use that forecasts multi-decade risk; (5) individualized MHT decision-making based on timing, route, symptoms, and vascular risk; and (6) structured lifestyle interventions supported by digital tools. Pragmatic trials, specialized education, and implementation science can ensure scalable, equitable impact.

ARTICLE INFORMATION

Affiliations

Division of Neurology, Department of Medicine, University of Toronto, ON, Canada (A.P.). JS Sonshine Centre, University Health Network, Toronto, ON, Canada (A.P., S.I., E.K.). Krembil Brain Institute, University Health Network, Toronto, ON, Canada (A.P.). Institute of Health Policy, Management, and Evaluation, Dalla Lana School of Public Health, University of Toronto, ON, Canada (E.K.). Brain Care Labs, Massachusetts General Hospital (E.R., S.D.S.). Department of Neurology, Mass General Brigham, MA (E.R., S.D.S.). Broad Institute of MIT and Harvard, MA (E.R., S.D.S.). Center for Genomic Medicine, Massachusetts General Hospital, MA (E.R., S.D.S.). Department of Neurology and Neurosurgery, Hospital Moinhos de Vento, Porto Alegre, Brazil (A.C.S.).

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